

Rejection Sampling Notes

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Problem Statement

We want to generate a sample from $\text{Gamma}(2, 1)$, where $X \sim \text{Gamma}(2, 1)$ is conditioned on $X \geq 5$. Thus, we define:

$$X^* = (X \mid X \geq 5)$$

Our target distribution is:

$$f(x) = \frac{xe^{-x}}{6e^{-5}}$$

Our initial proposal probability density function is $\text{Exp}(\lambda = 1)$, denoted as $g(x)$:

$$g(x) = e^{-x}$$

Finding the Scaling Factor M

We need a scaling factor M such that:

$$M \cdot g(x) \geq f(x) \quad \text{for } x \geq 5$$

We observe that for both $f(x)$ and $g(x)$, for all x_1, x_2 in the support of $f(x)$ and $g(x)$, if $x_2 > x_1$, then $f(x_1) > f(x_2)$ and $g(x_1) > g(x_2)$. Therefore, the maximum of the ratio $\frac{f(x)}{g(x)}$ occurs at the boundary $x = 5$.

At $x = 5$, we calculate M :

$$M = \frac{f(5)}{g(5)} = \frac{\left(\frac{5e^{-5}}{6e^{-5}}\right)}{e^{-5}}$$

Simplifying the expression:

$$M = \frac{5/6}{e^{-5}} = \frac{5}{6e^{-5}} \approx 123.6776$$